Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.12_k-1)

Metric

$$ds^2 = x_3^2 \sinh(x_1)^2 \, \mathrm{d}x_2 \otimes \mathrm{d}x_2 + x_3^2 \mathrm{d}x_1 \otimes \mathrm{d}x_1 + \left(\frac{9\, x_3^4}{\Lambda^2 x_3^8 + 12\, \Lambda m x_3^5 + 6\, \Lambda x_3^6 + 3\,(2\,\Lambda e^2 + 3) x_3^4 + 36\, m x_3^3 + 18\,(2\, m^2 + e^2) x_3^2 + 36\, m x_3 e^2 + 9\, e^4} \right) \mathrm{d}x_3 \otimes \mathrm{d}x_3 + \left(-\frac{\Lambda^2 x_3^8 + 12\, \Lambda m x_3^5 + 6\, \Lambda x_3^6 + 3\left(2\,\Lambda e^2 + 3\right) x_3^4 + 36\, m x_3^3 + 18\left(2\, m^2 + e^2\right) x_3^2 + 36\, m x_3 e^2 + 9\, e^4}{9\, x_3^4} \right) \mathrm{d}x_4 \otimes \mathrm{d}x_4$$

Coordinates

$$(x^\mu) = (x_2, x_1, x_3, x_4)$$

Metric Functions

$$\Sigma = \sinh(x_1)$$

Null Tetrad

Covariant components

$$l = \left(-\frac{3\sqrt{2}}{2\left(\Lambda x_3^2 + \frac{6\, m}{x_3} + \frac{3\, e^2}{x_3^2} + 3\right)} \right) dx_3 + \left(-\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2} m}{x_3} - \frac{\sqrt{2} e^2}{2\, x_3^2} \right) dx_4$$

$$n = \left(\frac{3\sqrt{2}}{2\left(\Lambda x_3^2 + \frac{6\, m}{x_3} + \frac{3\, e^2}{x_3^2} + 3\right)} \right) dx_3 + \left(-\frac{1}{6}\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\sqrt{2} - \frac{\sqrt{2} m}{x_3} - \frac{\sqrt{2} e^2}{2\, x_3^2} \right) dx_4$$

$$m = \left(\frac{1}{2}\sqrt{2} x_3 \sinh(x_1) \right) dx_2 + \left(\frac{1}{2} i \sqrt{2} x_3 \right) dx_1$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}x_3 \sinh\left(x_1\right)\right)\,dx_2 + \left(-\frac{1}{2}i\,\sqrt{2}x_3\right)\,dx_1$$

Contravariant components

$$l = \left(-\frac{1}{6}\,\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\,\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2\,x_3^2}\right)\,dx_3 + \left(\frac{3\,x_3^2}{\sqrt{2}\Lambda x_3^4 + 6\,\sqrt{2}m x_3 + 3\,\sqrt{2}x_3^2 + 3\,\sqrt{2}e^2}\right)\,dx_4$$

$$n = \left(\frac{1}{6}\,\sqrt{2}\Lambda x_3^2 + \frac{1}{2}\,\sqrt{2} + \frac{\sqrt{2}m}{x_3} + \frac{\sqrt{2}e^2}{2\,x_3^2}\right)\,dx_3 + \left(\frac{3\,x_3^2}{\sqrt{2}\Lambda x_3^4 + 6\,\sqrt{2}m x_3 + 3\,\sqrt{2}x_3^2 + 3\,\sqrt{2}e^2}\right)\,dx_4$$

$$m = \left(\frac{\sqrt{2}}{2\,x_3 \sinh\left(x_1\right)}\right)\,dx_2 + \left(\frac{i\,\sqrt{2}}{2\,x_3}\right)\,dx_1$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,x_3 \sinh\left(x_1\right)}\right)\,dx_2 + \left(-\frac{i\,\sqrt{2}}{2\,x_3}\right)\,dx_1$$

Directional Derivatives

$$D = -\frac{1}{6}\,\sqrt{2}\Lambda x_3^2 - \frac{1}{2}\,\sqrt{2} - \frac{\sqrt{2}m}{x_3} - \frac{\sqrt{2}e^2}{2\,x_3^2}\,\partial_{x_3} + \frac{3\,x_3^2}{\sqrt{2}\Lambda x_3^4 + 6\,\sqrt{2}m x_3 + 3\,\sqrt{2}x_3^2 + 3\,\sqrt{2}e^2}\,\partial_{x_4}$$

$$\Delta = \frac{1}{6}\,\sqrt{2}\Lambda x_3^2 + \frac{1}{2}\,\sqrt{2} + \frac{\sqrt{2}m}{x_3} + \frac{\sqrt{2}e^2}{2\,x_3^2}\,\partial_{x_3} + \frac{3\,x_3^2}{\sqrt{2}\Lambda x_3^4 + 6\,\sqrt{2}m x_3 + 3\,\sqrt{2}x_3^2 + 3\,\sqrt{2}e^2}\,\partial_{x_4}$$

$$\delta = \frac{\sqrt{2}}{2\,x_3 \sinh\left(x_1\right)}\,\partial_{x_2} + \frac{i\,\sqrt{2}}{2\,x_3}\,\partial_{x_1}$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,x_3 \sinh\left(x_1\right)}\,\partial_{x_2} - \frac{i\,\sqrt{2}}{2\,x_3}\,\partial_{x_1}$$

Spin Coefficients

$$\rho = \frac{1}{6}\sqrt{2}\Lambda x_3 + \frac{\sqrt{2}m}{x_3^2} + \frac{\sqrt{2}}{2\,x_3} + \frac{\sqrt{2}e^2}{2\,x_3^3}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{6}\sqrt{2}\Lambda x_3 + \frac{\sqrt{2}m}{x_3^2} + \frac{\sqrt{2}}{2\,x_3} + \frac{\sqrt{2}e^2}{2\,x_3^3}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{2}\cosh\left(x_1\right)}{4\,x_3\sinh\left(x_1\right)}$$

$$\beta = \frac{i\sqrt{2}\cosh\left(x_1\right)}{4\,x_3\sinh\left(x_1\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{6}\sqrt{2}\Lambda x_3 + \frac{\sqrt{2}m}{2\,x_3^2} + \frac{\sqrt{2}e^2}{2\,x_3^3}$$

$$\gamma = -\frac{1}{6}\sqrt{2}\Lambda x_3 + \frac{\sqrt{2}m}{2\,x_3^2} + \frac{\sqrt{2}e^2}{2\,x_3^3}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{5}{36}\Lambda^2 x_3^2 - \frac{\Lambda m}{3\,x_3} - \frac{\Lambda e^2}{6\,x_3^2} + \frac{2\,m^2}{x_3^4} - \frac{1}{2\,x_3^2} + \frac{5\,me^2}{x_3^5} + \frac{e^2}{x_3^4} + \frac{9\,e^4}{4\,x_3^6}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{5}{36}\Lambda^2 x_3^2 - \frac{1}{3}\Lambda - \frac{\Lambda m}{3\,x_3} - \frac{\Lambda e^2}{18\,x_3^2} - \frac{1}{6\,x_3^2} - \frac{me^2}{3\,x_3^5} - \frac{e^4}{4\,x_3^6}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = \frac{1}{18}\,\Lambda^2 x_3^2 + \frac{\Lambda e^2}{9\,x_3^2} + \frac{4\,m^2}{x_3^4} + \frac{2\,m}{x_3^3} + \frac{1}{3\,x_3^2} + \frac{20\,me^2}{3\,x_3^5} + \frac{2\,e^2}{x_3^4} + \frac{5\,e^4}{2\,x_3^6}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"